



An automated approach for hemocytometer cell counting based on image-processing method

Yu-Wei Chen^a, Pei-Ju Chiang^{b,*}

^a Department of Mechanical Engineering, National Chung Cheng University, Chia-Yi, Taiwan

^b Department of System and Naval Mechatronic Engineering, National Cheng Kung University, Tainan City, Taiwan

ARTICLE INFO

Keywords:

Automated cell counting
Human cervical carcinoma
Viability analysis
Hemocytometer
Image processing

ABSTRACT

Evaluating cell viability plays an important role in routine cell culture, toxicology, and drug development. Trypan blue stain is one of the simplest assays that are used to determine the number of viable cells in a cell suspension. The stained cell suspension can then be counted under a microscope using a hemocytometer. Hemocytometers have a fixed volume, making it possible to calculate the concentration of cells in the sample and measure viability. Most of the related studies are conducted using manual counting methods. However, manual counting is time-consuming and laborious. While various automated cell counting devices have been proposed, they tend to be too expensive for widespread deployment. Although low-cost automated cell counting methods presented in the literatures provide high accuracy and cost-effective alternatives, only cells in the field of view of microscope with unknown volume are counted. This will make the concentration inestimable. Accordingly, the present study proposes a low-cost but accurate method for hemocytometer cell counting using an image-processing approach. The accuracy of the proposed method is demonstrated by comparing the cell counting results obtained for human cervical carcinoma (HeLa) cells with the manual counts. It is shown that the proposed approach has a mean absolute percentage error of less than 6.26 % for samples containing a mixture of live and dead cells. Moreover, the proposed system is equipped with a simple and intuitive human machine interface such that the counting task can be performed by individuals with no prior experience in the image-processing field.

1. Introduction

Determining the concentration of cells is essential for many cell culturing and biological applications. Furthermore, evaluating cell viability plays a fundamental role in assessing the effectiveness of different substances in mitigating the effects of toxic materials [1] or gauging the success of different cryopreservation techniques [2,3]. Cell viability is generally quantified by counting the number of healthy cells in a sample [4]. One of the most common methods for determining cell viability is the trypan blue dye exclusion assay, in which the dead cells are selectively colored blue such that the live and dead cells can be easily distinguished and counted manually under a microscope using a hemocytometer [5]. Hemocytometers have specified grids and volumes to ensure that the cells are uniformly distributed and the same volume is analyzed each time, thereby allowing for accurate volumetric counts. Most biological laboratories have microscopes coupled with digital cameras such that cells can be visualized and analyzed through the captured images. Hence, integrating an existing microscope with a reusable hemocytometer provides a low-cost approach for performing

cell counting. However, manual cell counting is a tedious and time-consuming task. Moreover, the cell counting results may differ from one person to another due to the inherently subjective nature of the counting task.

Accordingly, various automated counting devices have been proposed, including flow cytometers and Coulter counters. In flow cytometers [6,7], the cells are first combined with certain fluorescent substances based on the cell characteristics such that they exhibit fluorescence. They are then dispersed in a buffer solution and placed in a flow cytometer, where they are excited by a laser source and emit an optical signal. Finally, the signal is filtered and converted to an equivalent electrical signal, from which the number of cells is then automatically determined. In fact, flow cytometers have many features; however, they are usually expensive, not solely for cell counting, and also complex to use. In Coulter counters [8], the cells are suspended within a given volume of fluid, and the number of cells is counted simply by recording the variation in the electric current as the fluid passes through one or more micro-channels of the device. However, Coulter counters are also expensive and cannot differentiate between live and

* Corresponding author.

<https://doi.org/10.1016/j.measurement.2024.114894>

Received 22 February 2024; Received in revised form 8 May 2024; Accepted 10 May 2024

Available online 11 May 2024

0263-2241/© 2024 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

dead cells.

To address the limitations of existing cell counting methods, various image-processing schemes have been proposed based on ImageJ [9], Metamorph, Cellprofiler [10] or CellC [11] software. However, these methods still rely on users to observe changes in the images and to determine various parameters in advance, such as the threshold value to be used in separating the cells from the background. As a result, the reliability of these methods again depends heavily on the subjective judgment of the user. Accordingly, many researchers have proposed image processing-based methods capable of counting or analyzing specific cells in a fully-automated manner. For example, Kachouie et al. [12] proposed a mask-based system in which the cells in a microchip were identified and counted by performing correlation tests on the locations of each pixel in the image. Militello et al. [13] proposed a fully automated method to calculate the percentage of area-based cell colony for the surviving fraction quantification. Considering that the area covered by colony is proportional to the colony number and size as well as to the growth rate, this measurement relies just on the colony covering percentage and does not consider the colony number to prevent inconsistencies due to intra- and inter-operator variability. Maula et al. [14] present AutoCellSeg that implements a supervised automatic and robust image segmentation method. The methods used in AutoCellSeg are based on a priori feature extraction for parameters tuning and post-processing. A priori knowledge about the size, number, shape and other features of the colonies should be defined by the user. Rundo et al. [15] present an automated cell detection and counting method for activity detection of fluorescent labeled cell nuclei in time-lapse microscopy. Sergioli et al. [16] introduced a multidisciplinary effort which involves image processing, machine learning, quantum information theory, cell biology and Helstrom quantum classifier to the binary classification of colonies and background area over four different cell lines. Chiang et al. [17] proposed a fully automated colony counting system using image processing methods to estimate the number of viable bacteria in a known volume of suspension. Since only surviving microbes grow and form colonies on the plate, by counting the number of colonies, the number of viable bacteria can be obtained. Ruberto et al. [18] presented an automated system for detecting and classifying malaria parasites in images of Giemsa-stained blood slides based on morphological operators. De Cevalrho et al. [19] proposed a method for counting yeast cells using a morphological segmentation technique. Quelhas et al. [20] used a sliding band filter (SBF) to differentiate and separate the nuclei and cytoplasm of drosophila cells. Loukas et al. [21] presented an image analysis-based method for counting cancer cell nuclei in tissue sections using a combined principal component analysis (PCA) and histogram thresholding technique. Han et al. [22] proposed a support vector machine (SVM) classification system for automated cell counting and tissue architecture analysis based on Laplace edge feature detection. Xu et al. [23] proposed an object detector based on YOLOv4 to detect blood cell objects such as red blood cells, white blood cells and platelets. Aung et al. [24,25] proposed an automatic counting for live and dead cells from trypan blue-stained images by image analysis based on adaptive K-means clustering, thresholding and morphological operation techniques. However, although the live cell counting had high accuracy (>95 %), the dead cell counting had a poor correlation coefficient with the manual counting ($R^2 < 0.7$) and low accuracy (<80 %). Besides, the counting of cells on the grid lines was not clearly defined. Kuijpers et al. [26] present a machine learning (ML) model based on YOLOv4 that is able to perform cell counts with a high accuracy for Trypan blue-stained insect cells. Akkoyun [27] proposes an automated cell counting software using computer vision technology and experimental investigation for automated cell and bacterium counting. The software processes images for calculating cell/bacterium count, concerning pre-defined user parameters.

As described above, the literature contains many proposals for automated cell identification and counting using image processing-based techniques. However, very few image-processing methods have

been proposed for automated cell counting using conventional hemocytometer devices, one of the least expensive and most widely applied instruments for cell counting. Thus, the present study proposes an automatic image processing-based method for counting live and dead cells in hemocytometer images which simplifies the counting task and eliminates the uncertainty caused by the subjective assessment of different individuals.

In general, a hemocytometer contains two counting chambers, each etched with a grid of perpendicular lines. The area covered by the grid lines is specified in such a way that the cell concentration can be obtained simply by counting the number of cells within each square. In practice, however, only the corner squares are used for cell counting purposes. Thus, in the image-processing method proposed in this study, the valid counting areas are first extracted from the hemocytometer image, and the live and dead cells within these areas are then identified based on their corresponding intensity characteristics. In practical hemocytometer samples, some of the cells (both live and dead) may aggregate together to form clusters. To improve the counting performance, these clusters should be separated such that the cells can be individually counted. The literature contains various methods for achieving this using image processing-based techniques, including the search for concave points method [28,29], the calculation of arc-chord ratios method [30], the Hough transformation method [31–34], and watershed methods [35–37]. Among these methods, watershed methods are the most commonly used due to their inherent speed and simplicity. However, the direct application of watershed algorithms generally leads to an over-segmentation of the cells as a result of noise. Thus, in the present study, the number of cells in the hemocytometer image is determined using a combined approach based on both a watershed method and the intensity characteristics of the cells.

While various automated cell counting devices have been proposed, they tend to be too expensive for widespread deployment. Although low-cost automated cell counting methods presented in the literatures provide high accuracy and cost-effective alternatives, only cells in the field of view of microscope with unknown volume are counted. This will make the concentration inestimable and the statistical results unreliable since the cells may distribute non-uniformly. Accordingly, the present study proposes a low-cost but accurate method for hemocytometer cell counting using an image-processing approach. Cells in the counting region including the ones on the outer frames and grid lines are counted in the same way as the counting rules of hemocytometer, so the volume of the suspension and concentration can be clearly defined. The proposed method has the ability to distinguish and count both live and dead cells, and can also count cells in clusters with an acceptable accuracy. The accuracy of the proposed method is demonstrated by comparing the cell counting results obtained for human cervical carcinoma (HeLa) cells with the manual counts. It is shown that the counting results obtained using the proposed system are in good agreement with those obtained manually. Furthermore, a user-friendly graphical user interface (GUI) is developed and made freely available for download, providing researchers in the biomedicine field with a convenient instrument for the enumeration of cells.

The remainder of this paper is organized as follows. Section 2 reviews the basic principles and operating procedures of a conventional hemocytometer. Section 3 describes the image-processing technique proposed in the present study. Section 4 presents and discusses the experimental results and describes the proposed GUI. Finally, Section 5 provides some brief concluding remarks and indicates the intended direction of future research.

2. Operating principle of hemocytometer

Prior to injection in the hemocytometer, the cell solution is generally stained with trypan blue dye in order to distinguish the live cells from the dead cells. In particular, the dead cells become dark blue in color due to ruptures in their membranes, while the live cells, for which the

membranes are intact, remain transparent.

As described in the previous section, a hemocytometer contains two chambers, where each chamber is partitioned into nine squares with 1-mm^2 area, as shown in Fig. 1(a). The chambers have a height of 0.1 mm , and hence the volume of each square is equal to 0.1 mm^3 . When using a hemocytometer, the cell concentration of the injected solution is determined by counting only the number of cells in the corner squares N. However, to ensure the accuracy of the counting results, the cells should be uniformly distributed in the hemocytometer chambers. Thus, if the difference in the total number of cells in the corner squares of the two chambers (labeled as N in Fig. 1(a)) exceeds 20 % of their mean value, the hemocytometer is washed out and a new sample is injected in its place.

As shown in Fig. 1(b), the frames of the corner squares comprise three separate lines. For each square, the central line serves as the boundary, and the cells which touch these lines to the left or the top are counted, while those which touch the lines to the right or the bottom are not. Thus, in Fig. 1(b), the empty and solid circles denote cells which are and are not counted, respectively. The cell concentration of the injected solution is then determined as follows:

$$\text{Cell concentration} = \text{average cell counts per square} \times 10 \times 1000 \times \text{Dilution} \quad (\text{cells/mL}) \quad (1)$$

where *dilution* is the ratio of the diluted sample volume to the original sample volume. Since the volume of each square is equal to 0.1 mm^3 , multiplying the average number of cells in each square by 10 gives the average number of cells per 1 mm^3 of solution. Furthermore, multiplying the average cell number by 1000 gives the average number of cells per 1 mL of solution. The cell concentration of the original suspension (cells/mL) can then be obtained simply by multiplying the number of cells by the dilution ratio.

3. Materials and methods

To obtain hemocytometer images, 5 mL of human cervical carcinoma (HeLa) cells, the most widely-used human cell line in scientific research, was mixed with 5 mL of trypan blue dye. The solution was injected into a hemocytometer using a micropipette, and images were obtained through a $10\times$ ocular lens and a $10\times$ objective lens by a charge-coupled device (CCD) camera with a resolution of 1360×1024 pixels and an exposure time of $1/5.5$ sec. The captured images were pre-processed to compensate for the effects of non-uniform illumination and skewness, respectively, and the cells in each corner square (both live and dead) were then extracted using a combined watershed and cell intensity analysis method. Fig. 2 shows the block diagram depicting the image processing pipeline of proposed cell counting approach. The accuracy of the proposed approach was evaluated by comparing the achieved

counting results with respect to manual counting results. The details of the proposed image-processing method are described in the sections which follow.

3.1. Image pre-processing

3.1.1. Illumination correction and counting region extraction

Fig. 3(a) shows one original image of counting regions of the hemocytometer. A linear stretch operation modifying the brightness (intensity) values of pixels in the image linearly for contrast enhancement was applied to the color CCD image of each corner square (N). To reduce the computational load, Principal Component Analysis (PCA) [38] was then applied to convert the color image to a grayscale equivalent, as shown in Fig. 3(b). PCA is a mathematical technique that can effectively reduce the dimension of image data while still maintaining the principal properties of the original image. It generally consists of four major steps: (1) normalize image data; (2) calculate covariance matrix from the image data; (3) perform Single Value Decomposition (SVD); (4) find the projection of image data to the new basis with reduced features.

To correct for the effects of non-uniform illumination, the intensities of a certain number of points in the background region of the image were acquired in order to estimate the intensity variation over the image background [39,40]. The background points were selected automatically by partitioning the grayscale image into squares with a width twice as wide as the HeLa cell diameter ($20\text{--}40\text{ }\mu\text{m}$ depending on the culture conditions.). In the hemocytometer image, the background is darker than the live cells, but brighter than the dead cells. Thus, the intensities of the pixels within each square were ranked and the median value was taken as the background intensity of the square. In such a way, a list of intensity values $f(x_i, y_i)$ and coordinates (x_i, y_i) of points in the background can be acquired. A polynomial function using all of the acquired background intensity values and their coordinates can be applied to define the background brightness as follows:

$$f(x_i, y_i) = a_0 + a_1x_i + a_2y_i + a_3x_i^2 + a_4y_i^2 + a_5x_iy_i \quad (2)$$

where i indicates the index of acquired background points, and $a_0, a_1, \dots, a_5$ are model parameters to be determined. With n background points, n polynomial equations can be obtained to form a matrix–vector equation as follows:

$$\begin{bmatrix} f(x_1, y_1) \\ f(x_2, y_2) \\ \vdots \\ f(x_n, y_n) \end{bmatrix} = \begin{bmatrix} 1 & x_1 & y_1 & x_1^2 & y_1^2 & x_1y_1 \\ 1 & x_2 & y_2 & x_2^2 & y_2^2 & x_2y_2 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 1 & x_n & y_n & x_n^2 & y_n^2 & x_ny_n \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \\ a_3 \\ a_4 \\ a_5 \end{bmatrix} \quad (3)$$

A least-squares fitting technique was then applied to obtain the model parameters $a_0, a_1, \dots, a_5$ to fit the background surface. The shadow image was then obtained by subtracting the mean intensity value from the fitted image. Finally, the effects of non-uniform illumination were eliminated by subtracting the shadow image from the original PCA image. To extract the counting regions of the CCD image, the intensity-corrected image was first binarized using Otsu's method [41] to produce the result shown in Fig. 3(c). The Otsu's method calculates the best threshold to binarize the image into the foreground and background components based on the histogram of the intensity values so that the between class variance (i.e., separability between the foreground and background) is maximized.

As shown in Fig. 3(c), each counting region consisted mainly of 16 squares, i.e., the 16 largest black areas in the image surrounded by white grid lines. However, after the binarization process, some of the grid lines may not be well connected, and hence some neighboring squares become inter-connected and form a single region. As a result, some regions of the image outside of the 16 squares may be erroneously extracted, as shown in Fig. 4(a). To resolve this problem, the dark spots

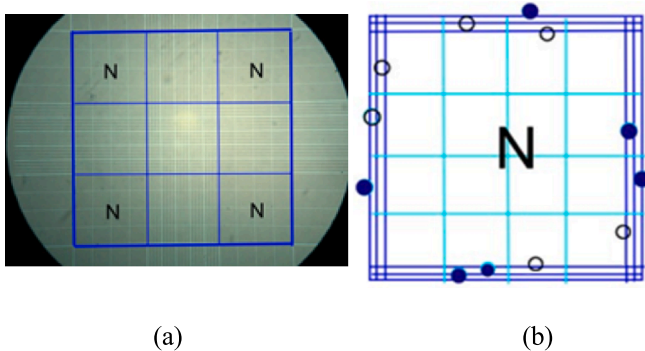


Fig. 1. (a) Hemocytometer chamber under $10\times$ objective lens. (b) Counting rules for hemocytometer. Note that only the empty circles are counted.

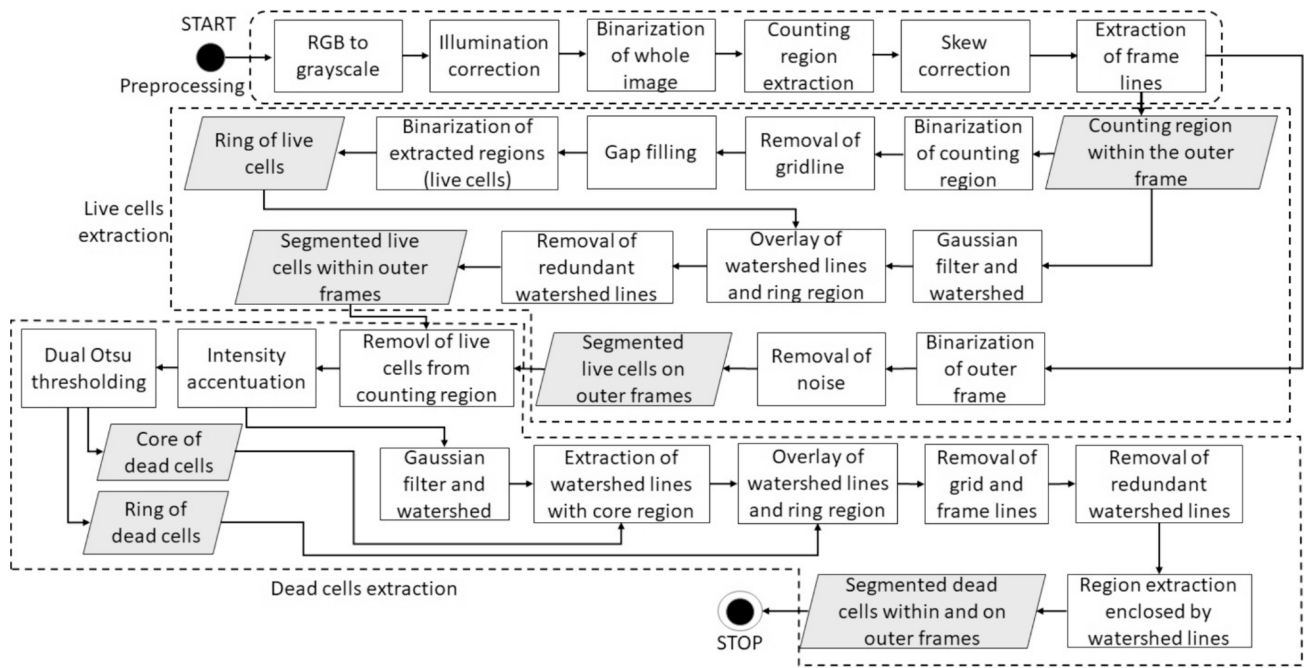


Fig. 2. Block diagram depicting the image processing pipeline of proposed cell counting approach.

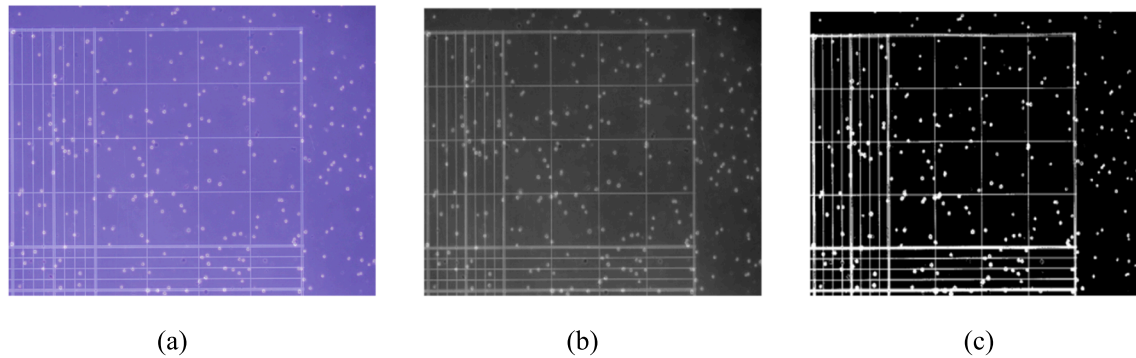


Fig. 3. (a) Original color image of counting regions. (b) Grayscale image of counting regions. (c) Binarized image of intensity-corrected image.

in the image were eliminated using a hole-filling operation (fills in shapes of black pixels surrounded by white pixels) (Fig. 4(b)) and the counting region, N , was then obtained by dilating the resulting image with a flat square structuring element (SE) with a side length equal to the width of the outer frame. In such a way, the white regions were dilated and the black regions were eroded. The black grid lines with width less than the outer frame were eliminated, and the 16 squares were connected, as shown in Fig. 4(c). The connected component with the largest area in Fig. 4(c) was selected and superimposed on the image in Fig. 4(b) using an AND logical operator to obtain the result shown in Fig. 4(d). Fig. 4(d) was then dilated using a square with a side length equal to twice the width of the outer frame. Finally, the approximate location of the counting region was obtained by subtracting Fig. 4(d) from the dilated image to produce the result shown in Fig. 4(e).

3.1.2. Skew correction

As described in Section 2, the numbers of cells in the corner squares are counted. The frames of the corner squares comprise three different lines (as shown in Fig. 1(b)). For each square, the second line serves as the boundary of the counting area. If the image is skewed, it is difficult to extract the boundary. Therefore, to ensure the accurate extraction of each counting region, the CCD image was corrected for skewness. In

general, when an image is skewed, the distance between the highest point in the image and the lowest point increases. Thus, to correct for potential skew in the present CCD imaging process, each captured image was rotated from -5° to $+5^\circ$ in increments of 0.1° , and the distance between the highest and lowest points in the image was calculated in each case. The image with the smallest distance was then taken as the skew-corrected image and retained for further image processing analysis.

3.1.3. Extraction of frame lines from counting regions

As described earlier, in processing hemocytometer images, the cells touching the central lines of the frame to the left or the top are counted, while those touching the central lines to the right or the bottom are not. To facilitate the automatic counting process, the frame lines of each counting region must therefore be located. As shown in Fig. 5(a), each counting region contains five vertical lines, i.e., two outer frame lines and three internal grid lines. Furthermore, for each vertical line, there are two edges, where black and white alternate. In the image-processing method proposed in this study, the location of each vertical line was determined by recording the horizontal coordinates of the pixels along each edge and then taking the coordinate value which most frequently occurred as the horizontal coordinate of the edge. A similar approach

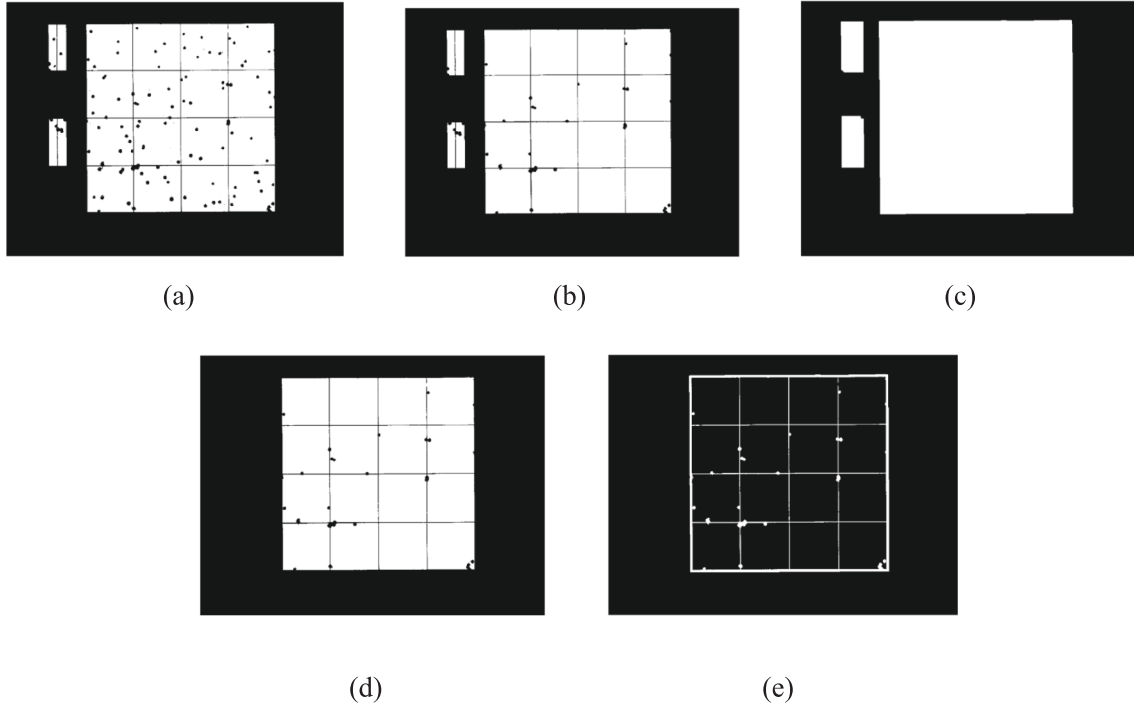


Fig. 4. (a) 16 largest connected components before extraction. (b) Hole filling of (a). (c) Dilation of (b). (d) Intersection of largest connected component of (b) with (c). (e) Results obtained by subtracting (d) from dilated image of (d) obtained using square SE with side length equal to twice outer frame width.

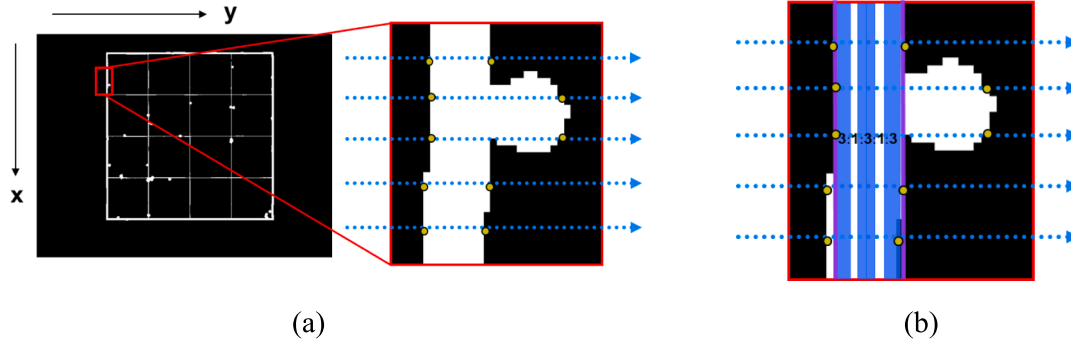


Fig. 5. (a) Enlarged view of outer frame of Fig. 3(e). (b) Three lines on the outer frame. Note that the width ratio of the three lines and intervals is equal to 3:1:3:1:3.

was applied to the edge of each horizontal line in the image in order to determine the locations of the upper and lower frames and horizontal grid lines, respectively.

As shown in Fig. 1(b), the outer frame of each corner frame consists of three lines separated by a distance equal to approximately one third of the line width. A 3:1:3:1:3 ratio was thus applied to separate the frame, as shown in Fig. 5(b), in which the three lines are marked in blue while the intervals between them are marked in white. The entire frame lines and grid lines were thus reconstructed. The counting regions between the left and top outer frames and the central lines to the right and bottom were then extracted, as shown in Fig. 6(a).

3.2. Extraction of live cells

3.2.1. Extraction of live cells within outer frame

To extract the live cells within the outer frame, a window with a width equal to half the cell diameter was first used to perform the average filtering of Fig. 6(a) in order to eliminate noise. Otsu's global thresholding method [41] was then employed to binarize the image, as shown in Fig. 6(b). In addition to live cells, the extracted image also

includes elements of the frames. Consequently, an XOR logical operator was applied to Fig. 6(b) and the image of the reconstructed frame in order to remove the frame components. The corresponding results are shown in Fig. 6(c). The removal process caused some cells touching the frame to be cut in two. Thus, to repair these cells, the areas of the remaining connected components in Fig. 6(c) which were originally linked to the grid lines were computed and, for each component, if the area was less than half the area of a cell, the remaining region was dilated using a flat square SE with a side length equal to the frame width. An AND logical operator was then applied to the dilated image and the reconstructed frame lines in order to obtain the areas of the regions bridging the gaps between the cells. These regions were then added to Fig. 6(c) in order to fill the gaps in the broken cells (see Fig. 6(d)). To count the cells within the outer frame, the cells touching the outer frame were screened out using a morphological border-clearing algorithm [42] which simply extracts the objects touching the border and then obtains the new image with no objects touching the borders. Finally, a flat disc SE with a width equal to half the radius of the cell was used to perform an opening operation on Fig. 6(d) in order to remove any noise with a size less than half the cell radius. The opening operation erodes an image

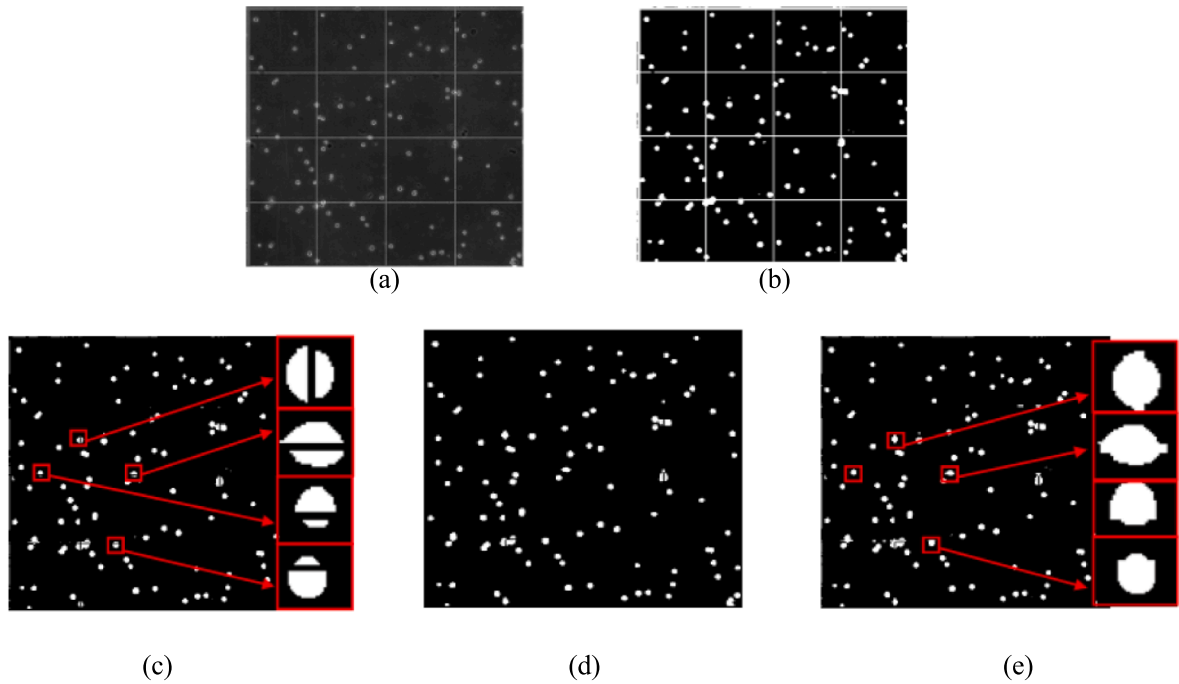


Fig. 6. (a) Extraction of counting area in Fig. 2 with skew correction. (b) Binarized image of (a). (c) Results of grid line removal from (b). (d) Results of gap filling in (c). (e) Results after removal of noise and cells on outer frame.

and then dilates the eroded image, and is useful for removing small objects and thin lines from an image while preserving the shape and size of larger objects in the image. Fig. 6(e) shows the final extraction results. Once live cells within the outer frame were extracted, the separation methods in Section 3.2.3 was applied to separate clustered live cells.

3.2.2. Extraction of live cells on outer frame

Fig. 7(a) shows an enlarged image of a live cell located on the outer frame. It is seen that the outer ring of the cell is brighter than the three lines comprising the frame. Thus, an intensity value located between that of the outer ring of the cell and that of the outer frame was used as a threshold to segment the cell from the frame. To determine the appropriate threshold value, Fig. 6(e) was placed as a mask over Fig. 6(a) prior to the binarization process in order to determine the location of the cell's outer ring. The Otsu threshold was calculated based on the histogram of the intensity values of the outer ring and the three frame lines, and was then applied to the region of outer frame to segment the cell from the frame. Since cells lining up along the outer frame is rarely seen, to reduce the false positive error (noise incorrectly indicates live cells), in the segmentation process, any extracted components with a size less than half the cell size or greater than twice the cell diameter were

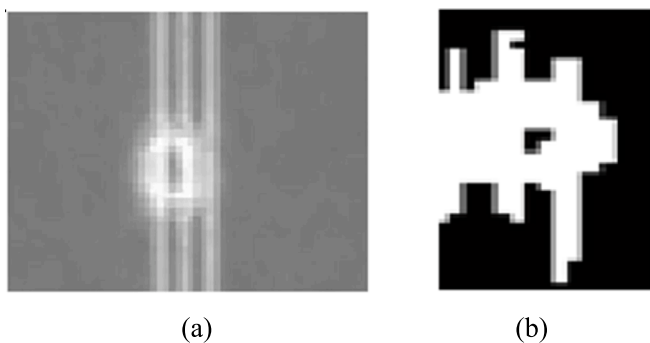


Fig. 7. (a) Grayscale image of live cell on outer frame. (b) Enlarged image of live cell extracted from outer frame.

considered to be noise and were removed accordingly. Finally, any cells touching the central lines to the right or bottom were eliminated, thereby completing the extraction process. Fig. 7(b) shows the typical extraction result obtained for a live cell on the outer frame.

3.2.3. Separation of clustered live cells

Once the cell colonies were extracted from the counting regions, a watershed method was employed to separate any clustered cells. The watershed algorithm divides an image into segments using topographic information. It treats the image as a topographic surface, and local minima are marked as starting points (holes). The entire topography is flooded from below by letting water rise through the holes at a uniform rate. When the rising water in distinct catchment basins is about to merge, a dam is built to prevent the merging. The flooding will eventually reach a stage when only the tops of the dams are visible above the water line. These dam boundaries (watershed lines) correspond to the divide lines of the watersheds. Therefore, they are the boundaries extracted by a watershed segmentation algorithm. To prevent excessive segmentation caused by the presence of too many local minima, the pre-processed image shown in Fig. 6(a) was first filtered using a Gaussian filter with a standard deviation equal to one sixth of the cell radius. Fig. 6(e) were then used as masks to derive the images of the live cell colonies. Although the Gaussian filter mitigated the excessive segmentation problem during the watershed process, a small portion of the cell colonies were still overly segmented, as shown in Fig. 8. Nonetheless, most of the watershed lines were accurately constructed on the bright ring regions of the cells, i.e., only a minority of the lines lay in the background region. The intensity values of the bright cell rings are much higher than those of the background. Thus, the bright cell rings were extracted through a binarization operation, and the number of pixels corresponding to the watershed lines located on the bright rings was computed. If the number was less than 60 % of the total number of pixels constituting the watershed line, the line was eliminated. Finally, the number of live cells was computed by counting the number of connected components enclosed within the remaining watershed lines.

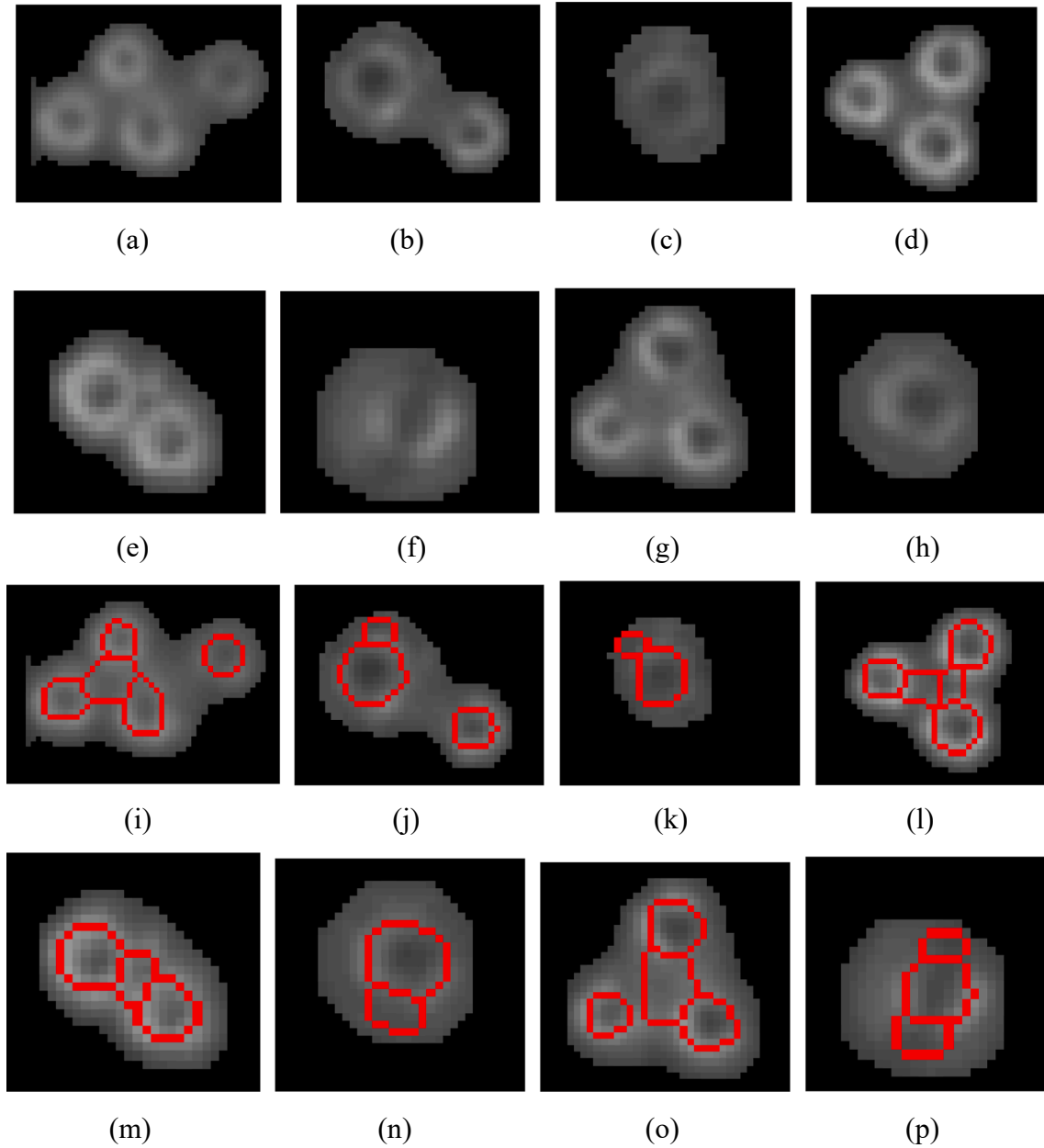


Fig. 8. (a)–(h) Enlarged images of extracted live cells. (i)–(p) Overlay of watershed lines on images shown in (a)–(h).

3.3. Extraction of dead cells

3.3.1. Intensity accentuation of dead cells

Fig. 9(a) presents an enlarged image of a typical dead cell. It is seen that the pixels within the cell have a low intensity, whereas those located in the outer ring have a high intensity. Since the intensity values within the cells are inconsistent and close to that of the background, extracting the dead cells presents a major challenge. Accordingly, in the image-processing method proposed in the present study, the extraction process was facilitated using an operational mask, as described in the following.

It is seen in Fig. 9(a) that the diameter of the dark region in the center of the cell is equal to approximately one fourth the diameter of the complete cell. The operational mask was therefore divided into two regions of a corresponding size, as shown by the black and white circles in Fig. 9(b), respectively. In particular, the overall size of the mask was set equal to the diameter of the cell such that the diameter of the white region was equal to one fourth that of the black region. In the extraction

process, the mask was moved in raster order pixel-by-pixel over the entire image, with the average intensity of the pixels in the black region being subtracted from the average intensity of the pixels in the white region in every step. When the center of the mask was located at the center of the dead cell, the average intensity of the white region reduced since it contained the darkest region of the cell. By contrast, the average intensity of the black region increased since it encompassed the high-intensity region of the outer ring. Thus, the subtraction process yielded a large negative output value. As the center of the mask moved away from the center of the cell, the area of the darker region within the white circle decreased, leading to an increased average intensity within the white region. By contrast, the number of background pixels within the black region increased, resulting in a reduction in the average intensity within the black region. Consequently, the subtraction process resulted in a smaller negative value. When the center of the mask lay on the outer ring of the cell, the subtraction process produced a positive value since the white region encompassed the high-intensity region of the outer ring while the black region encompassed the background region and low-

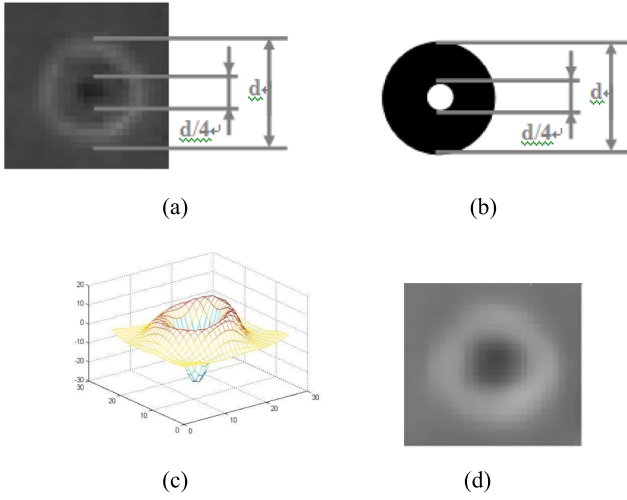


Fig. 9. (a) Enlarged image of dead cell, where d indicates approximate cell diameter. (b) Mask used for intensity accentuation of dead cells. (c) Results of applying mask in (b) to (a). (d) Image of (c) after scaling values to full intensity range.

intensity region of the cell center. Finally, when the mask was located entirely in the background region of the image, the subtraction process produced a value close to zero. Fig. 9(c) shows the typical subtraction values obtained for a single dead cell. Fig. 9(d) shows the resulting image obtained by scaling the subtraction values to the full intensity range. It is seen that the intensities of the pixels in the center of the dead cell, the outer ring, and background region, respectively, are more

distinguishable than those in Fig. 9(a). Note that to prevent the intensity values of the frame and live cells from affecting the calculation process, any pixels located on the frame or live cells obtained previously were excluded from the segmentation process. Fig. 10(a) shows the typical results obtained when applying the mask to an image containing dead cells.

3.3.2. Extraction of dead cells not on frame or grid lines

The image shown in Fig. 10(a) contains three different intensity ranges corresponding to the dark center of the dead cells, the bright rings surrounding the dead cells, and the background region, respectively. The dual Otsu threshold segmentation method [41] was thus used to calculate two threshold values and segment Fig. 10(a) into three corresponding regions. As mentioned earlier, the Otsu's method can calculate the best threshold so that the separability between each class is maximized. As described in the previous section, to prevent the intensity values of the frame and live cells from affecting the intensity calculation results, any live cells or pixels located on the frame obtained previously were excluded in the segmentation process. Fig. 10(b) shows the segmentation results obtained for pixels with intensity values lower than the lower Otsu threshold, i.e., the pixels in the center of the dead cells. By contrast, Fig. 10(c) shows the segmentation results for the pixels with intensity values greater than the higher Otsu threshold. To identify all of the regions containing dead cells, and to segment the overlapping dead cells, Fig. 10(a) was filtered using a Gaussian filter with a standard deviation equal to one sixth of the cell radius and was then processed using the watershed algorithm. The corresponding results are presented in Fig. 10(d). It is seen that the cells are overly segmented. Since Fig. 10(b) mainly shows the centers of dead cells, Fig. 10(d) was superimposed on it, and only those watershed lines surrounding the white regions were considered as outlines of dead cells. Fig. 10(e) shows the regions

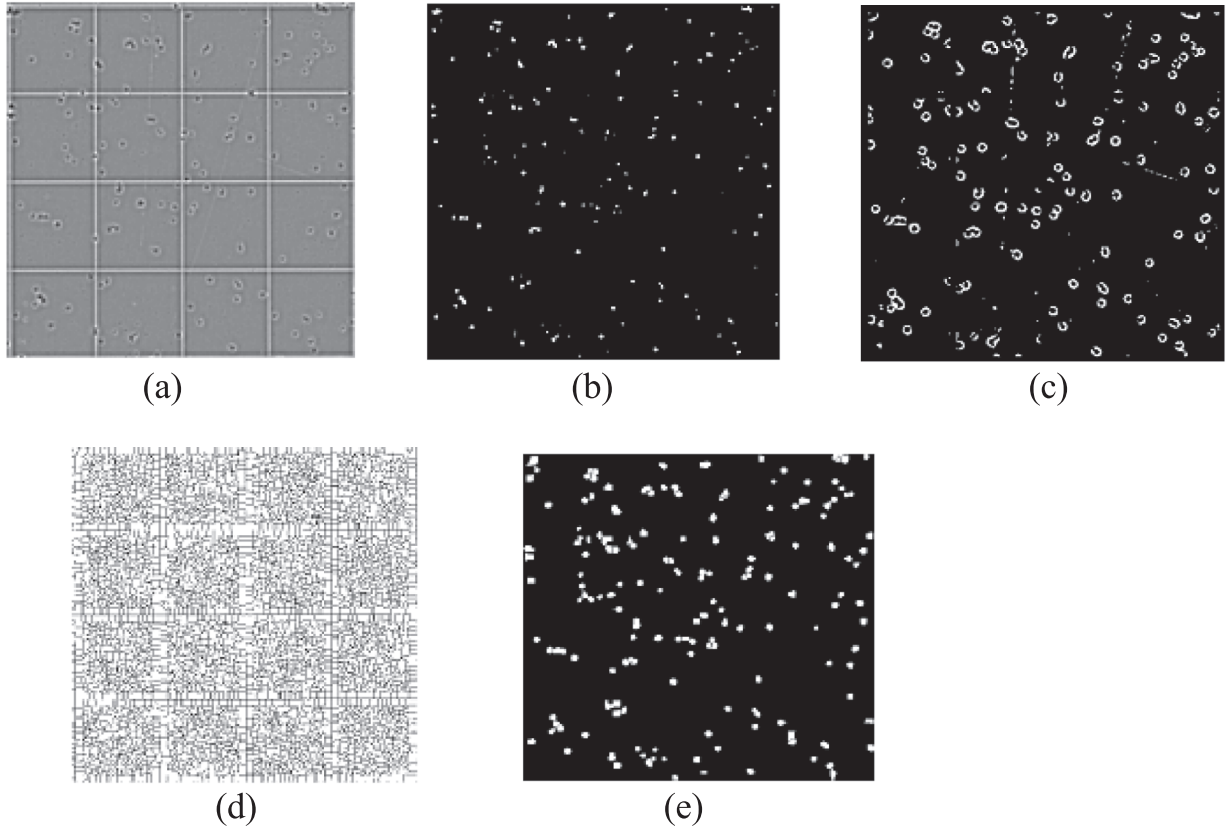


Fig. 10. (a) Mask operation results for an image containing dead cells. (b) Extraction of pixels in (a) with intensity values less than lower value of dual Otsu threshold. (c) Extraction of pixels in (a) with intensity values greater than higher value of dual Otsu threshold. (d) Results of (a) after application of Gaussian filter and watershed algorithm. (e) Extraction of regions enclosed in watershed lines of (d) surrounding white regions in (b).

enclosed by the elected watershed lines. It is seen that most of the white regions in Fig. 10(e) are dead cells. Those regions not belonging to dead cells were excluded by exploiting the ring characteristic of dead cells. In particular, Fig. 10(c) mainly shows the rings of the dead cells, and was thus overlaid with the elected watershed lines. Fig. 11(a) and (b) shows an enlarged image of a dead cell and the overlaid watershed lines, respectively. It is observed that most of the elected watershed lines fall on the white rings. For each enclosed watershed line, after eliminating the pixels connected with other basins, if more than 50 % of the remaining pixels fell on the white region, the region enclosed by the watershed line was taken to be a dead cell; otherwise, the watershed line was regarded as noise and was removed accordingly. The remaining watershed lines are shown in Fig. 11(c), in which the enclosed regions represent dead cells.

3.3.3. Extraction of dead cells on frame or grid lines

For cells touching the frame or grid lines (see Fig. 12(a)), removing the dilated grid lines also results in a partial removal of the extracted ring (see Fig. 12(b)). To properly extract the dead cells, the watershed lines within the dilated frame lines were thus first eliminated, as shown in Fig. 12(c). The proportion of the remaining watershed lines falling on the white regions was then calculated. If the value was greater than 50 %, the regions enclosed by the corresponding watershed lines prior to elimination (see Fig. 12(d)) were considered as dead cells. Finally, any cells touching the central lines of the frame to the right or bottom were eliminated. The number of dead cells lying on the frame or grid lines was thus obtained. This number was added to the number of dead cells not on the frame or grid lines obtained in the previous section, yielding the

Negative cases. Table 1 shows the four possible situations for the live cell identification outcome, where True-Positive represents the number of actual live cells correctly identified as live cells by the image-processing system; False-Positive represents the number of non-live-cells incorrectly identified as live cells; False-Negative represents the number of live cells incorrectly identified as non-live-cells; and True-Negative represents the number of non-live-cells correctly identified as non-live-cells.

The precision of the proposed system (also known as the positive predictive value) and the recall (also known as the sensitivity) were evaluated respectively as [43]

$$\begin{aligned} \text{Precision} &= \frac{\text{Number of lived cells retrieved}}{\text{Total number of components retrieved}} \\ &= \frac{\text{True positive}}{\text{True positive} + \text{False positive}} \end{aligned} \quad (4)$$

$$\begin{aligned} \text{Recall} &= \frac{\text{Number of lived cells retrieved}}{\text{Total number of existing live cells}} \\ &= \frac{\text{True positive}}{\text{True positive} + \text{False negative}} \end{aligned} \quad (5)$$

Furthermore, the F-measure, which combines the precision characteristic and recall characteristic of the system, was defined as [43]

$$F\text{-measure} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (6)$$

Finally, the absolute percentage error (APE) was calculated as:

$$\text{APE} (\%) = \frac{100 \times |\text{Number of lived cells retrieved} - \text{Number of existing live cells}|}{\text{Number of existing live cells}} \quad (7)$$

total number of dead cells in the image.

4. Results and discussion

The cell counting performance of the proposed image-processing method was evaluated using 24 images containing live cells only, 24 images containing dead cells only, and 12 images containing both live and dead cells. Each image contained between 49 and 247 cells. For each image, the counting results were compared with those obtained manually by an experienced technician. The manual counting results were taken as the benchmark in evaluating the True and False cases, while the results obtained using the automated image-processing method were taken as the cell identification outcome in evaluating the Positive and

Note that the evaluation measures given in Eqs. (4)–(7) were also applied to the dead cells.

Tables 2–4 present the statistical results for the precision, recall, F-measure and APE of the proposed system for the images containing live cells only, dead cells only, and a mixture of live and dead cells, respectively. It is seen that the mean values of the precision, recall and F-measure are mostly greater than 0.95. Furthermore, the APE for images containing live cells only and dead cells only are 2.63 % and 3.92 %, respectively, while those for the images containing a mixture of live and dead cells are 4.59 % for live cells and 6.26 % for dead cells.

It is noted that the proposed counting system has a lower accuracy for dead cells than for live cells in all cases. This performance deficit

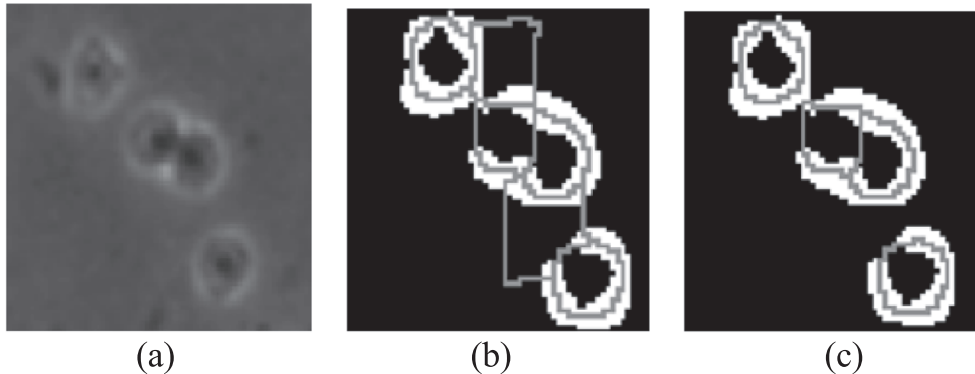


Fig. 11. (a) Enlarged image of dead cells. (b) Overlay of elected watershed lines on (a). (c) Results after removal of watershed lines with less than 50% of stand-alone pixels (i.e., not connecting with other basins) falling on white regions in (b).

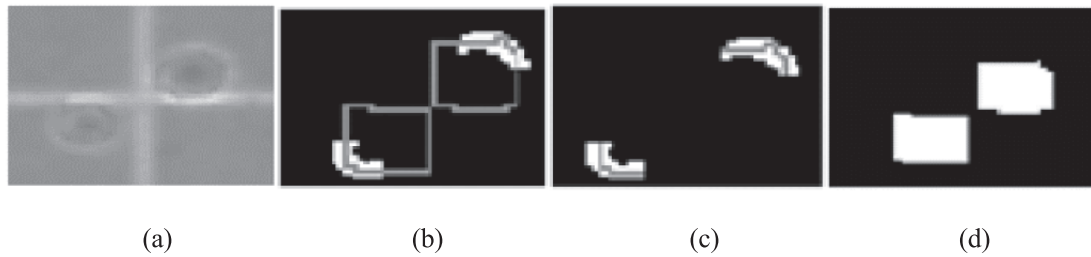


Fig. 12. Extraction of dead cells on frame lines. (a) Grayscale image of dead cells touching frame lines. (b) Ring extraction results after overlay of watershed lines and removal of dilated frame lines. (c) Results after removal of watershed lines falling within dilated frame lines in (b). (d) Regions enclosed in watershed lines in (b).

Table 1

Confusion matrix for live cells.

		Manual counting	
		True	False
Automated counting	Positive	True positive (correct result)	False positive (unexpected result)
	Negative	False negative (missing result)	True negative (correct absence)

Table 2

Statistical analysis results for images containing only live cells.

Evaluation measures (live cells)	Live cells			
	Min.	Max.	Std.	Avg.
Precision	0.93	1	0.02	0.98
Recall	0.92	0.99	0.02	0.97
F-measure	0.95	0.99	0.01	0.97
APE	0 %	7.06 %	2.15 %	2.63 %

Table 3

Statistical analysis results for images containing only dead cells.

Evaluation measures (dead cells)	Dead cells			
	Min.	Max.	Std.	Avg.
Precision	0.90	0.99	0.03	0.95
Recall	0.93	0.99	0.02	0.96
F-measure	0.94	0.97	0.01	0.96
APE	0.71 %	10.66 %	2.94 %	3.92 %

stems mainly from the miscellaneous collection of noise associated with the dead cells (e.g., dead cell fragments). Generally, the size of such noise is small and can be eliminated by the Gaussian filter. However, for congregated noise with a size similar to that of the cells, the system may erroneously identify the noise as a cell. Thus, the performance of the system can be improved by avoiding the generation of dead cell fragments during the culturing process.

Figs. 13 and 14 show the correlations between the automated counts and manual counts for the images containing single cells and a mixture of cells, respectively. It is seen that a strong correlation exists between the two counts in every case. Figs. 15 and 16 show the Bland-Altman plots of the automated counts and manual counts for the images containing single cells and a mixture of cells, respectively. It is seen that the average difference values between automated counts and manual counts

for images containing single and mixture of live and dead cells are both less than one. And 95 % of the differences between the automated counts and manual counts are less than ± 9 and ± 5 for images containing single and mixture of live and dead cells, respectively. Thus, the accuracy of the proposed automated system is further confirmed.

Table 5 compares the counting performance of the proposed image processing-based method with that of two commercial cell counters and ImageJ Macros [46], respectively. In all four cases, the counting process is performed using the images acquired through a microscope. For example, the TC20 automated cell counter [44] uses an auto-focus microscope which analyzes multiple focal planes to identify the best plane and then uses the image acquired at the best focal plane to calculate the total cell count and assess the cell viability. The Cellometer Mini [45] utilizes bright field imaging and pattern-recognition software to identify and count individual live cells and dead cells stained with trypan blue. Cellometer Mini can also analyze individual cells in clusters. ImageJ Macros [46] provides a complete solution for automatic cell counting, in which a conventional light microscope is equipped with a web camera to obtain images of a suspension of cells and the cells are then counted automatically using an image-processing algorithm based on the ImageJ toolbox. Compared to commercial cell counters, ImageJ Macros is free for download. However, it cannot analyze the cell viability or count individual cells within clusters. By contrast, the method proposed in this study has the ability to distinguish and count both live and dead cells and can also count cells in clusters with an acceptable accuracy.

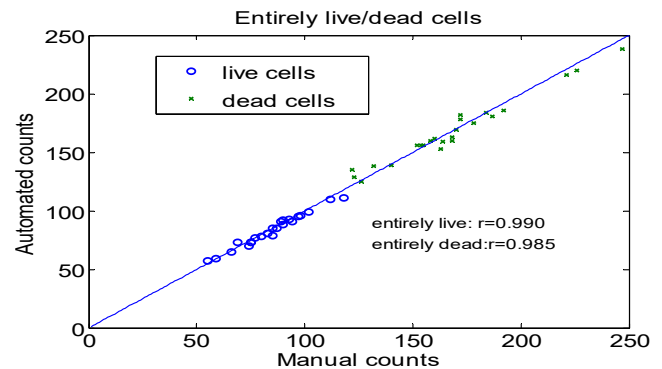


Fig. 13. Correlation of automated counts and manual counts for images containing live cells only or dead cells only.

Table 4

Statistical analysis results for images containing mixture of live and dead cells.

Evaluation measures (mixed cells)	Live cells				Dead cells			
	Min.	Max.	Std.	Avg.	Min.	Max.	Std.	Avg.
Precision	1	1	0	1	0.86	0.98	0.039	0.91
Recall	0.86	1	0.04	0.95	0.89	1	0.03	0.95
F-measure	0.92	1	0.02	0.98	0.90	0.96	0.02	0.93
APE	0 %	14.19 %	4.12 %	4.59 %	0 %	16.67 %	5.90 %	6.26 %

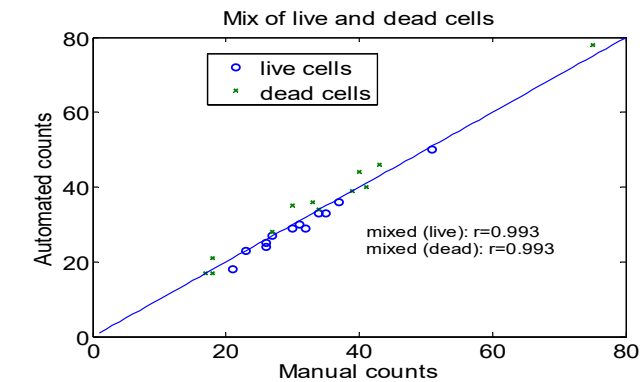


Fig. 14. Correlation of automated counts and manual counts for images containing mixture of live and dead cells.

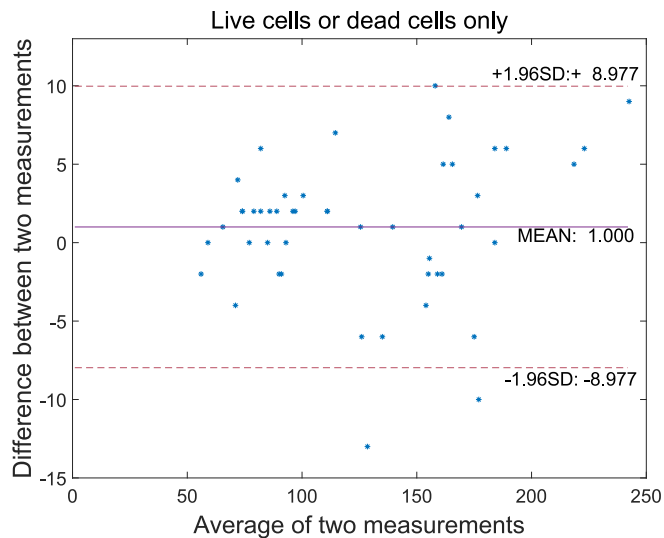


Fig. 15. Bland-Altman plot of automated counts and manual counts for images containing live cells only or dead cells only.

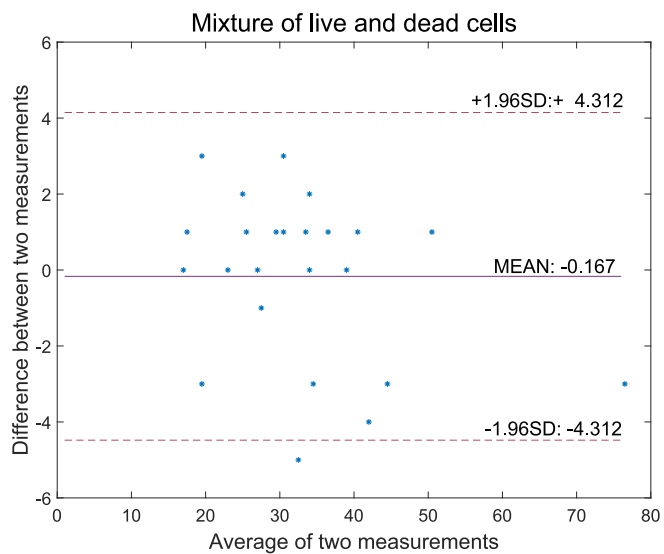


Fig. 16. Bland-Altman plot of automated counts and manual counts for images containing mixture of live and dead cells.

Table 5

The comparison of commercial cell counters, ImageJ Macro and proposed method.

Name	Biorad TC20 Automated Cell Counter[44]	Nexcelom Cellometer Mini[45]	ImageJ Macros[46]	Proposed method
Accuracy	Comparable to results obtained with a hemocytometer	Not available	Same as manual counting under the same conditions	Refer to Tables 2–4
Cell size (μm)	6–50	5–60	Not limited	Not limited
Functions	Cell counting	Cell counting	Cell counting	Cell counting
	Cell viability	Cell viability		Cell viability
		Cell declustering		Cell declustering
Price (USD)	2500	4000	0	0

Table 6

The comparison of cost-effective image-analysis-based cell counting methods and proposed method.

Name	Akkoyun [27]	Kuijpers [26]	Aung [24,25]	Proposed method
Accuracy	Single cell/ bacteria accuracy: 99 % Mixed cell/ bacteria accuracy: 98 %	Live cell F1 scores: 0.97 Dead cells F1 scores: 0.96	Live cell accuracy: 83 ± 6 % Dead cell accuracy: 63 ± 10 %	Live cells Precision: 0.98, Dead cell Precision: 0.95
Counting field	Field of view of microscope	Field of view of microscope	Counting area of hemocytometer	Counting area of hemocytometer
Source code	Not available	Not available	Not available	Available

In addition to commercial cell counters and ImageJ Macro, the literature contains many proposals for automated cell identification and counting using image processing-based techniques. Some image-processing methods have been proposed for automated cell counting using conventional hemocytometer devices. Table 6 is the comparison between these methods and proposed method. However, the comparison of the efficiency is not provided since the computing equipment is different from each other and not easy to compare. All of the methods are applicable to cells stained with trypan blue. Among these methods, Akkoyun [27] proposes an automated cell counting software using computer vision technology for automated cell/bacterium counting with pre-defined user parameters. Kuijpers et al. [26] present a machine learning (ML) model based on YOLOv4 that is able to perform cell counts with a high accuracy for Trypan blue-stained insect cells. Although these automated cell counting methods based on image analysis provides high accuracy and cost-effective solutions, only cells in the field of view of microscope with unknown volume are counted. This will make the concentration inestimable and the statistical results unreliable since the cells may distribute non-uniformly. Instead, in this study, cells in the counting region including the ones on the outer frames and grid lines are counted in the same way as the counting rules of hemocytometer, so the volume of the suspension and concentration can be clearly defined. Similar to proposed method, Aung et al. [24,25] presented an automatic counting for live and dead cells from trypan blue-stained images of hemocytometer by image analysis based on adaptive K-means clustering, thresholding and morphological operation techniques. However, although the live cell counting had high accuracy (>95 %), the dead cell

counting had a low accuracy ($<80\%$) and poor correlation coefficient with the manual counting ($R^2 < 0.7$). Besides, the counting of cells on the grid lines was not clearly defined. In contrast, the proposed method demonstrates high counting accuracy for both live cells and dead cells.

Additionally, all of these methods do not open the source codes to the public except the proposed method.

The study demonstrates the accuracy of the proposed method for HeLa cells. However, the proposed method can be applied to any cell lines (such as human breast cancer cell line, MDA-MB-231) with spherical shapes as Fig. 3(a) shown after stained with trypan blue, and is thus applicable to most biological cells in suspension culture. Nevertheless, based on the image processing pipeline of proposed counting approach, this method is only for cells stained with trypan blue in hemocytometer. In addition, the captured image must cover the entire corner square.

An intuitive graphical user interface (GUI) was developed for the proposed image-processing system such that the cell counting process can be performed by biomedicine practitioners with little or no prior experience in the image-processing field (see Fig. 17). To facilitate the automated counting process, the diameters of the live and dead cells, and the width of the outer frame, are first required. Accordingly, in the GUI, the user presses the “Select live cell” icon and then uses the cursor to draw a box around the area enclosing a single cell. Having done so, the system calculates the cell diameter automatically. Similarly, to calculate the diameter of the dead cells and the width of the outer frame, the user simply selects a representative dead cell and one horizontal segment of the outer frame, respectively. When the user then presses the “Start” command, the system proceeds to enumerate all the cells in the image automatically without any further user involvement. With an Intel i7-8700 CPU running at a 3.2 GHz clock speed, the enumeration is

typically completed within 35 s (depending on the number of cells). The enumeration results are presented on the screen, with the live and dead cells shown in red and green, respectively.

The GUI and code, together with the 60 test images used in the present study, are available for free download at <http://140.116.40.42:5000/sharing/iDRFWu2bf>. Notably, the image-processing method described in this study was implemented in MATLAB. However, it can also be implemented in other languages, such as C/C++. Biomedical laboratories are invariably equipped with microscopes for recording, detecting and analyzing the status of cells. By integrating these microscopes with the GUI proposed in this study, researchers in the biomedicine field gain a simple yet reliable tool for the low-cost and rapid enumeration of cells.

5. Conclusions and future work

Hemocytometers are inexpensive and widely used for cell counting. However, the manual counting process is laborious and time consuming. Accordingly, this study has presented an image processing-based system for counting the live and dead cells in hemocytometer images in an automated manner. The study has considered HeLa cells. However, the proposed method can be applied to any cell lines stained with trypan blue with a spherical shape, and is thus applicable to most biological cells in suspension culture. A user-friendly GUI has been developed to support the implementation of the proposed system. It has been shown that a strong correlation exists between the automated counts and the manual counts for samples containing only live or dead cells and samples containing a mixture of both live and dead cells. Thus, the proposed method provides an effective and practical solution for cell counting in hemocytometer images. Future studies will aim to further improve the

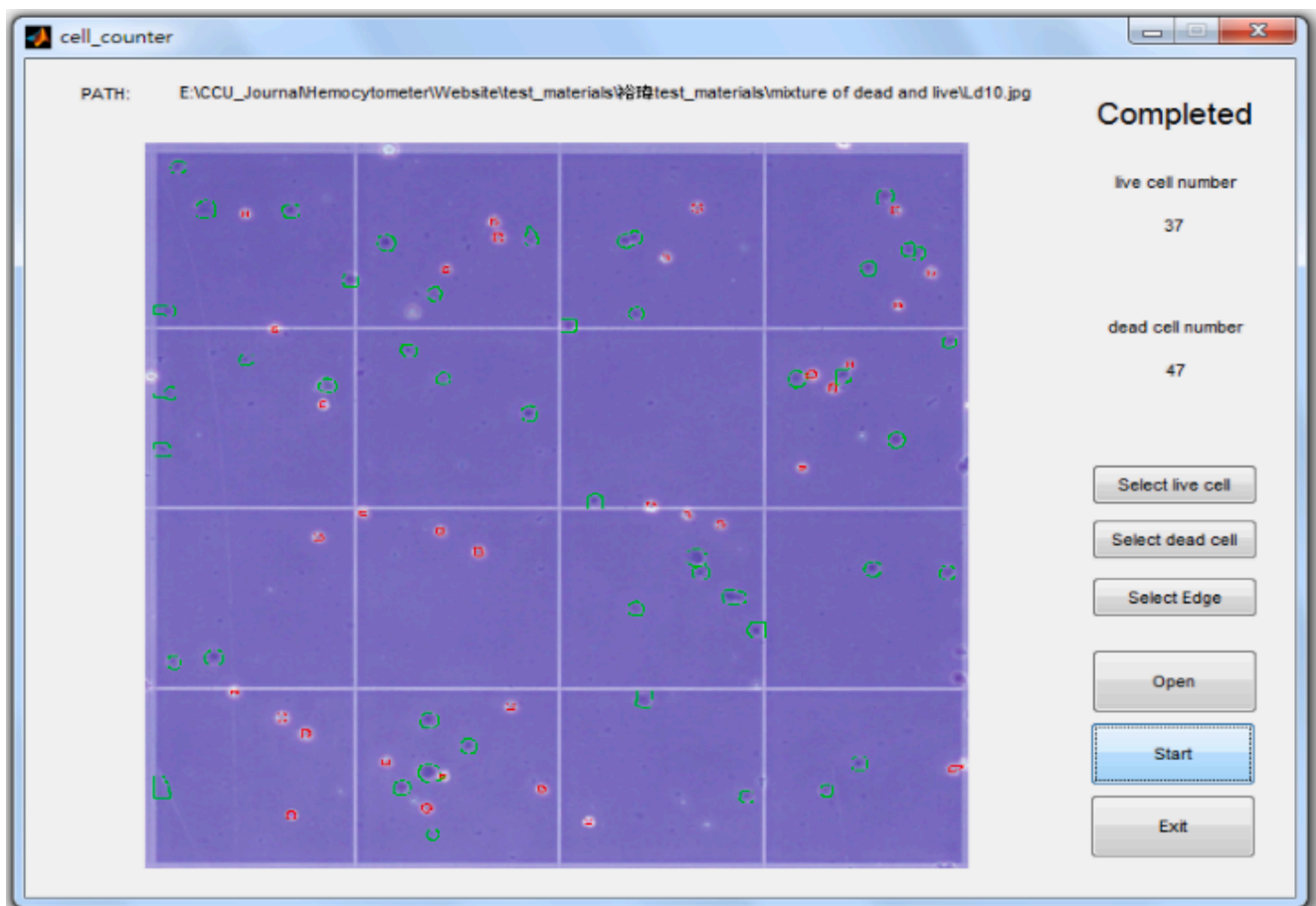


Fig. 17. Design of GUI.FIGURE.

efficiency of the proposed method. In addition, the system will be extended to enable the differentiation and counting of different cell lines in the same hemocytometer image. The study demonstrates the accuracy of the proposed method for Hela cells. However, the proposed method can be applied to any cell lines with similar morphology as Fig. 3(a) shown, and is thus applicable to most biological cells in suspension culture. Nevertheless, this method can be applied only to images with cells stained with trypan blue with spherical shapes in hemocytometer. In addition, the captured image must cover the entire corner square.

Funding

This research was supported in part by Higher Education Sprout Project, Ministry of Education to the Headquarters of University Advancement at National Cheng Kung University (NCKU), and funded by the National Science and Technology Council, Taiwan, R.O.C., under Grant No. NSTC 112-2221-E-006-142.

CRediT authorship contribution statement

Yu-Wei Chen: Writing – original draft, Visualization, Validation, Software, Methodology, Conceptualization. **Pei-Ju Chiang:** Writing – review & editing, Validation, Supervision, Project administration, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgments

The authors would like to thank Pro. Chun-Ping Jen of National Cheng Chung University, Mechanical Engineering, for use of their facilities, cell lines and images.

References

- [1] B. Lindner, U. Seydel, Mass spectrometric analysis of drug-induced changes in Na⁺ and K⁺ contents of single bacterial cells, *J. Gen. Microbiol.* 129 (1) (1983) 51–55.
- [2] Y. Pichugin, G.M. Fahy, R. Morin, Cryopreservation of rat hippocampal slices by vitrification, *Cryobiology* 52 (2) (2006) 228–240.
- [3] A.M. Crutchfield, K.R. Diller, J.J. Brand, Cryopreservation of *Chlamydomonas reinhardtii* (chlorophyta), *Eur. J. Phycol.* 34 (1) (1999) 43–52.
- [4] M.J. Stoddart, Cell viability assays: introduction, in: M.J. Stoddart (Ed.), *Mammalian Cell Viability: Methods and Protocols*, Humana Press, New York, 2011, pp. 1–6.
- [5] R. Ian Freshney, *Culture of Animal Cells: A Manual of Basic Technique and Specialized Applications*, Wiley-Liss, Wilmington, DE, 2005.
- [6] M.G. Macey, *Flow Cytometry: Principles and Applications*, Humana Press, New York, 2010.
- [7] A.L. Givan, *Flow Cytometry: First Principles*, Wiley-Liss, Wilmington, DE, 2001.
- [8] M. Don Graham, The coulter principle: foundation of an industry, *J. Assoc. Lab. Autom.* 8 (6) (2003) 72–81.
- [9] T.J. Collins, ImageJ for microscopy, *Biotechniques* 43 (S1) (2007) S25–S30.
- [10] A.E. Carpenter, et al., CellProfiler: image analysis software for identifying and quantifying cell phenotypes, *Genome Biology* 7 (10) (2006) R100.
- [11] J. Selinummi, J. Seppala, O. Yli-Harja, Software for quantification of labeled bacteria from digital microscope images by automated image analysis, *Biotechniques* 39 (6) (2005) 859–863.
- [12] N.N. Kachouie, L.F. Kang, A. Khademhosseini, Arraycount, an algorithm for automatic cell counting in microwell arrays, *Biotechniques* 47 (3) (2009) IX–XV.
- [13] C. Militello, L. Rundo, V. Conti, et al., Area-based cell colony surviving fraction evaluation: A novel fully automatic approach using general-purpose acquisition hardware, *Comput. Biol. Med.* 1 (89) (2017) 454–465.
- [14] A.M. Khan, A. Torelli, I. Wolf, G. Norbert, AutoCellSeg: robust automatic colony forming unit (CFU)/cell analysis using adaptive image segmentation and easy-to-use post-editing techniques, *Sci. Rep.* 8 (2018) 7302.
- [15] L. Rundo, A. Tangherloni, D.R. Tyson, et al., ACDC: Automated cell detection and counting for time-lapse fluorescence microscopy, *Appl. Sci. (Basel)* 10 (18) (2020) 6187.
- [16] G. Sergioli, C. Militello, L. Rundo, et al., A quantum-inspired classifier for clonogenic assay evaluations, *Sci. Rep.* 11 (2021) 2830.
- [17] P.-J. Chiang, M.-J. Tseng, Z.-S. He, C.-H. Li, Automated counting of bacterial colonies by image analysis, *J. Microbiol. Methods* 108 (2015) 74–82.
- [18] C.D. Ruberto, A. Dempster, S. Khan, B. Jarra, Analysis of infected blood cell images using morphological operators, *Image Vis. Comput.* 20 (2) (2002) 133–146.
- [19] M.A.G. de Carvalho, R. de A. Lotufo, M. Couprie, Morphological segmentation of yeast by image analysis, *Image Vis. Comput.* 25 (1) (2007) 34–39.
- [20] P. Quelhas, M. Marcuzzo, A.M. Mendonca, A. Campilho, Cell nuclei and cytoplasm joint segmentation using the sliding band filter, *IEEE Trans. Med. Imaging* 29 (8) (2010) 1463–1473.
- [21] C.G. Loukas, G.D. Wilson, B. Vojnovic, An image analysis-based approach for automated counting of cancer cell nuclei in tissue Sections, *Cytometry A* 55A (1) (2003) 30–42.
- [22] J.W. Han, T.P. Breckon, D.A. Randell, G. Landini, The application of support vector machine classification to detect cell nuclei for automated microscopy, *Mach. Vis. Appl.* 23 (1) (2012) 15–24.
- [23] F. Xu, X. Li, H. Yang, Y. Wang, W. Xiang, TE-YOLOF: Tiny and efficient YOLOF for blood cell detection, *Biomed. Signal. Process. Control* 73 (2022) 103416.
- [24] S.M. Aung, K. Kanokwiroon, T. Phairatana, S. Chatpun, Automatic counting for live and dead cells from trypan blue-stained images by image analysis based on adaptive k-means clustering, *J. Comput. Sci.* 15 (2019) 302–312.
- [25] S.M. Aung, K. Kanokwiroon, T. Phairatana, S. Chatpun, Live and dead cells counting from microscopic trypan blue staining images using thresholding and morphological operation techniques, *Int. J. Electr. Comput. Eng.* 9 (2019) 2460–2468.
- [26] L. Kuijpers, E. van Veen, L.A. van der Pol, N.H. Dekker, Automated cell counting for Trypan blue-stained cell cultures using machine learning, *PLoS One* 18 (11) (2023) e0291625.
- [27] F. Akkoyun, A. Özçelik, Rapid characterization of cell and bacteria counts using computer vision, *Turkish J. Nature Sci.* 10 (2021) 269–274.
- [28] X. Bai, C. Sun, F. Zhou, Splitting touching cells based on concave points and ellipse fitting, *Pattern Recogn.* 42 (11) (2009) 2434–2446.
- [29] G. Zhang, D.S. Jayas, N.D.G. White, Separation of touching grain kernels in an image by ellipse fitting algorithm, *Biosyst. Eng.* 92 (2) (2005) 135–142.
- [30] T. Huang, K. Lin, M. Li, C. Lu, A novel segmentation method for overlapping cells, *Intell. Control. Autom.* 2 (2006) 10486–10490.
- [31] T.J. Atherton, D.J. Kerbyson, The coherent circle Hough transform, in: *Proceedings of the British Machine Vision Conference*, Guildford, UK, 1993, pp. 269–278.
- [32] R.O. Duda, P.E. Hart, Use of the Hough transform to detect lines and curves in pictures, *Commun. ACM* 15 (1) (1972) 11–15.
- [33] C. de Solorzano, R. Malladi, S. Lelievre, S. Lockett, Segmentation of nuclei and cells using membrane related protein markers, *J. Microscopy-Oxf.* 201 (3) (2001) 404–415.
- [34] P.P. Guan, H. Yan, Blood cell image segmentation based on the Hough transform and fuzzy curve tracing, in: *Proceedings of 2011 International Conference on Machine Learning and Cybernetics (ICMLC)*, vol. 4, 2011, pp. 1696–1701.
- [35] Q. Pang, C. Yang, Y. Fan, Y. Chen, Overlapped cell image segmentation based on distance transform, *Intell. Control. Autom.* 2 (2006) 9858–9861.
- [36] L. Vincent, P. Soile, Watersheds in digital spaces: an efficient algorithm based on immersion simulations, *IEEE Trans. Pattern Anal. Mach. Intell.* 13 (6) (1991) 583–593.
- [37] L.E. Band, Topographic partition of watersheds with digital elevation models, *Water Resour. Res.* 22 (1) (1986) 15–24.
- [38] M. Petrou, P. Bosdogianni, *Image Processing – The Fundamentals*, Wiley & Sons Ltd., West Sussex, Chichester, UK, 1999, pp. 136–144.
- [39] J.C. Russ, *The Image Processing Handbook*, 2nd ed., IEEE Press, Boca Raton, Florid, 1995, pp. 184–187.
- [40] D. Tomazevic, B. Likar, F. Pernus, Comparative evaluation of retrospective shading correction methods, *J. Microsc.* 208 (3) (2002) 212–223.
- [41] N. Otsu, A threshold selection method from gray-level histograms, *IEEE Trans. Syst. Man Cybern.* 9 (1) (1979) 62–66.
- [42] R. Gonzales, R. Woods, *Digital Image Processing*, Prentice-Hall, Upper Saddle River, New Jersey, 2002.
- [43] S.V. Stehman, Selecting and interpreting measures of thematic classification accuracy, *Remote Sens. Environ.* 62 (1997) 77–89.
- [44] Bio-Rad, TC20 Automated Cell Counter, <https://www.bio-rad.com/en-tw/product/tc20-automated-cell-counter?ID=M7FBG34VY>.
- [45] Nexcelom, Cellometer Mini Automated Cell Counter, <https://www.nexcelom.com/nexcelom-products/automated-cell-counters/cellometer-mini-automated-cell-counter/>.
- [46] I.V. Grishagin, Automatic cell counting with ImageJ, *Anal. Biochem.* 473 (2015) 63–65.